

CHE 201 Spring 2020 Midterm 2

Thanh Nguyen

TOTAL POINTS

94 / 100

QUESTION 1

Question 1 25 pts

1.1 A 15 / 21

- 0 pts Correct
- 3 pts first one is incorrect
- 3 pts second one is incorrect
- 3 pts third one is incorrect
- ✓ - 3 pts fourth one is incorrect
- 3 pts fifth one is incorrect
- ✓ - 3 pts sixth one is incorrect
- 3 pts seventh one is incorrect

1.2 B 4 / 4

- ✓ - 0 pts Correct
- 4 pts not done or incorrect

QUESTION 2

Question 2 40 pts

2.1 a 10 / 10

- ✓ - 0 pts Correct
- 3 pts Qc not labeled or wrong direction
- 3 pts Qh not labeled or wrong direction
- 3 pts work not labeled or wrong direction

2.2 b 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect or not done
- 2 pts only one calculated
- 2 pts partially done/not calculated for year
- 2 pts without cooling is incorrect

2.3 C 5 / 5

- ✓ - 0 pts Correct
- 5 pts not done/incorrect

- 1 pts calculation error

2.4 d 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect/not done
- 3 pts Th is taken as 90C

2.5 e 10 / 10

- ✓ - 0 pts Correct
- 2 pts calculation error
- 10 pts not done/incorrect
- 5 pts only one calculated
- 2 pts unit conversion

2.6 f 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect /reverse formula/ not done
- 1 pts reversible/irreversible?
- 1 pts calculation error

QUESTION 3

Question 3 35 pts

3.1 a 5 / 5

- ✓ - 0 pts Correct
- 3 pts two outlets are not shown
- 2 pts inlet is not shown
- 1 pts boundary is not labeled

3.2 b 5 / 5

- ✓ - 0 pts Correct
- 2 pts open system not stated
- 2 pts steady state is not stated
- 1 pts delta PE is zero

3.3 C 5 / 5

✓ - 0 pts Correct

- 5 pts incorrect
- 2 pts work can not be canceled

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- 2 pts h_2 is wrong
- 2 pts KE term is not divided by 1000 (unit conversion)
- 10 pts incorrect or not done
- 2 pts not multiplied with mass outlets
- 1 pts work sign is wrong
- 2 pts work is not added
- 2 pts work unit is not converted
- 2 pts mass is wrong
- 2 pts h_1 is wrong
- 2 pts h_3 is wrong
- 0 pts [Click here to replace this description.](#)

calc error

3.5 e 5 / 5

✓ - 0 pts Correct

- 5 pts incorrect/not done
- 2 pts specific volume is wrong

3.6 f 5 / 5

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- 2 pts v_2 is wrong
- 2 pts state 1 not labeled
- 1 pts v_3 is wrong
- 5 pts not done /wrong
- 2 pts dome missing
- 1 pts state 3 is wrong
- 1 pts v_1 is wrong

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1.)

- A)
1. True
 2. True
 3. False
 4. True
 5. False
 6. False
 7. False

- B)
- Q in m^3/s Re is dimensionless
 D in m
 A in m^2

$$Re = \frac{QD}{\nu A} \rightarrow \nu = \frac{QD}{A Re}$$

$$\rightarrow \nu \text{ in } \frac{m^3}{s} \times m \times \frac{1}{m^2} = \frac{m^2}{s}$$

1.1 A 15 / 21

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- 3 pts third one is incorrect
- ✓ - 3 pts fourth one is incorrect
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1.)

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1.2 B 4 / 4

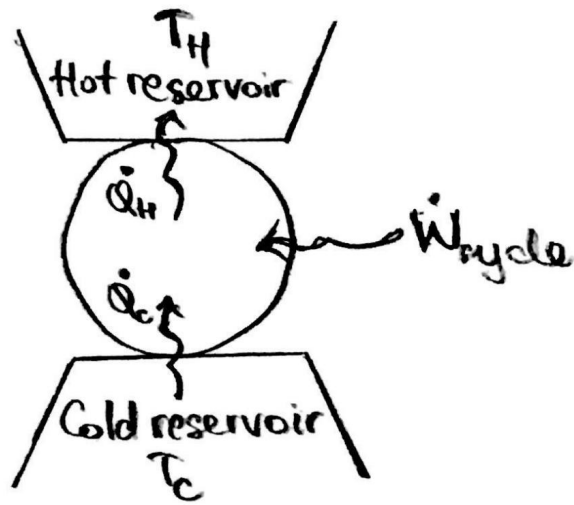
✓ - 0 pts Correct

- 4 pts not done or incorrect

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2.)

a.)



b.) Energy received by refrigeration: $Q_c = mc\Delta T$

• With cooling to room temperature

$$Q_{c1} = mc(T_{\text{kit}} - T_c) = (4\text{kg})(3.5\text{kJ/kg}\cdot^\circ\text{C})(25^\circ\text{C} - 4^\circ\text{C}) = 294\text{kJ per 1 dish}$$

$$Q_{c1} = \frac{294\text{kJ}}{1\text{dish}} \times \frac{3\text{dishes}}{1\text{week}} \times \frac{52\text{weeks}}{1\text{year}} \times 1\text{year} = 45864\text{kJ}$$

• Without cooling to room temperature

$$\frac{Q_{c2}}{Q_{c1}} = \frac{T_{\text{food}} - T_c}{T_{\text{kit}} - T_c} \rightarrow Q_{c2} = (45864\text{kJ}) \frac{90^\circ\text{C} - 4^\circ\text{C}}{25^\circ\text{C} - 4^\circ\text{C}} = 187824\text{kJ}$$

c.) Energy rejected by refrigeration: $Q_H =$

$$\beta = \frac{Q_c}{W} = \frac{Q_c}{Q_H - Q_c} \Rightarrow Q_H = \frac{1+\beta}{\beta} Q_c$$

• With cooling to room temperature

$$Q_{H1} = \frac{1+1.7}{1.7} Q_{c1} = \frac{2.7}{1.7} (45864\text{kJ}) = 72843\text{kJ}$$

• Without cooling to room temperature

$$Q_{H2} = \frac{1+1.7}{1.7} Q_{c2} = \frac{2.7}{1.7} (187824\text{kJ}) = 298308\text{kJ}$$

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d.) $\beta_{\max} = \frac{T_c}{T_H - T_c}$ where $T_H = T_{\text{kit}} = 25^\circ\text{C} + 273 = 298\text{K}$
 $T_c = T_{\text{re.}} = 4^\circ\text{C} + 273 = 277\text{K}$

$$\rightarrow \beta_{\max} = \frac{298\text{K}}{298\text{K} - 277\text{K}} = 14.2$$

e.) With cooling to room temperature

$$W_{\text{cycle 1}} = Q_{H1} - Q_{c1} = 72843\text{kJ} - 45864\text{kJ} = 26979\text{kJ}$$

Without cooling to room temperature

$$W_{\text{cycle 2}} = Q_{H2} - Q_{c2} = 298308\text{kJ} - 187824\text{kJ} = 110484\text{kJ}$$

→ Total amount of saving / year

$$(110484\text{kJ} - 26979\text{kJ}) \times \frac{1\text{KWs}}{1\text{kJ}} \times \frac{1\text{h}}{3600\text{s}} \times \$0.12/\text{KWh}$$
$$= \$2.78$$

$$\text{f.) } -\tilde{\sigma}_{\text{cycle}} = \frac{Q_{\text{in}}}{T_{\text{in}}} - \frac{Q_{\text{out}}}{T_{\text{out}}} = \frac{Q_c}{T_c} - \frac{Q_H}{T_H}$$
$$= \frac{45864\text{kJ}}{277\text{K}} - \frac{72843\text{kJ}}{298\text{K}} = -78.87\text{kJ/K}$$

→ $\tilde{\sigma}_{\text{cycle}} = 78.87\text{kJ/K} > 0 \rightarrow$ irreversible cycle.

2.1 a 10 / 10

✓ - 0 pts Correct

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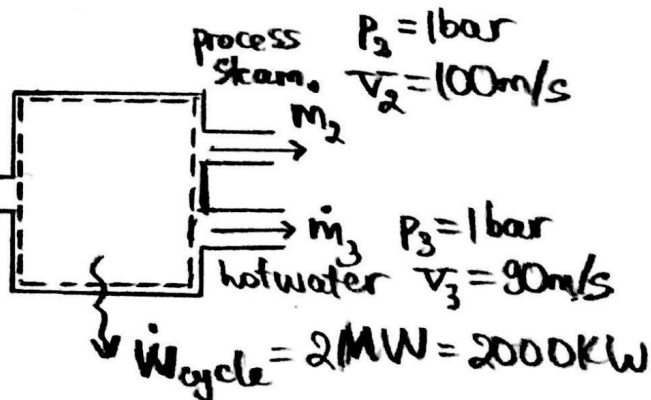
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3.)

a.) Sketch

vapor
 $T_1 = 400^\circ\text{C}$
 $P_1 = 30\text{ bar}$
 $V_1 = 40\text{ m/s}$
 $\dot{m}_i = 2\text{ kg/s}$



b.) Engineering Model

1. Control volume
2. Steady state
3. $\Delta PE = 0$
4. $\dot{m}_2 = \dot{m}_3 = \dot{m}_e$

c) Mass flow rate balance $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_i - \dot{m}_2 - \dot{m}_3 = 0$
 $\rightarrow \dot{m}_i = 2\dot{m}_e$

Energy rate balance $\frac{dE_{cv}}{dt} = 0$

$$\rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_1 + \frac{V_1^2}{2} + g z_1 \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right) = 0$$

$$\rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_1 + \frac{V_1^2}{2} \right) - \frac{\dot{m}_i}{2} \left(h_2 + h_3 + \frac{V_2^2 + V_3^2}{2} \right) = 0$$

$$\rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_1 - \frac{1}{2} (h_2 + h_3) + \frac{V_1^2}{2} - \frac{1}{2} \left(\frac{V_2^2 + V_3^2}{2} \right) \right) = 0$$

d) Table A-4, at $P_1 = 30\text{ bar}$, $T_1 = 400^\circ\text{C} \rightarrow h_1 = 3230.9\text{ kJ/kg}$

Table A-3, at $P_2 = 1\text{ bar} \rightarrow h_2 = 2675.5\text{ kJ/kg}$, $v_2 = 1.694\text{ m}^3/\text{kg}$, $T_2 = 99.63^\circ\text{C}$
 at $P_3 = 1\text{ bar} \rightarrow h_3 = 417.46\text{ kJ/kg}$, $v_3 = 1.0432 \times 10^{-3}\text{ m}^3/\text{kg}$, $T_3 = 99.63^\circ\text{C}$

$$\rightarrow \dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}_i \left(h_1 - \frac{1}{2} (h_2 + h_3) + \frac{V_1^2}{2} - \frac{1}{2} \left(\frac{V_2^2 + V_3^2}{2} \right) \right)$$

$$= 2000\text{ kW} - (2\text{ kg/s}) \left[(3230.9 - \frac{1}{2} (2675.5 + 417.46))\text{ kJ/kg} + \dots \right]$$

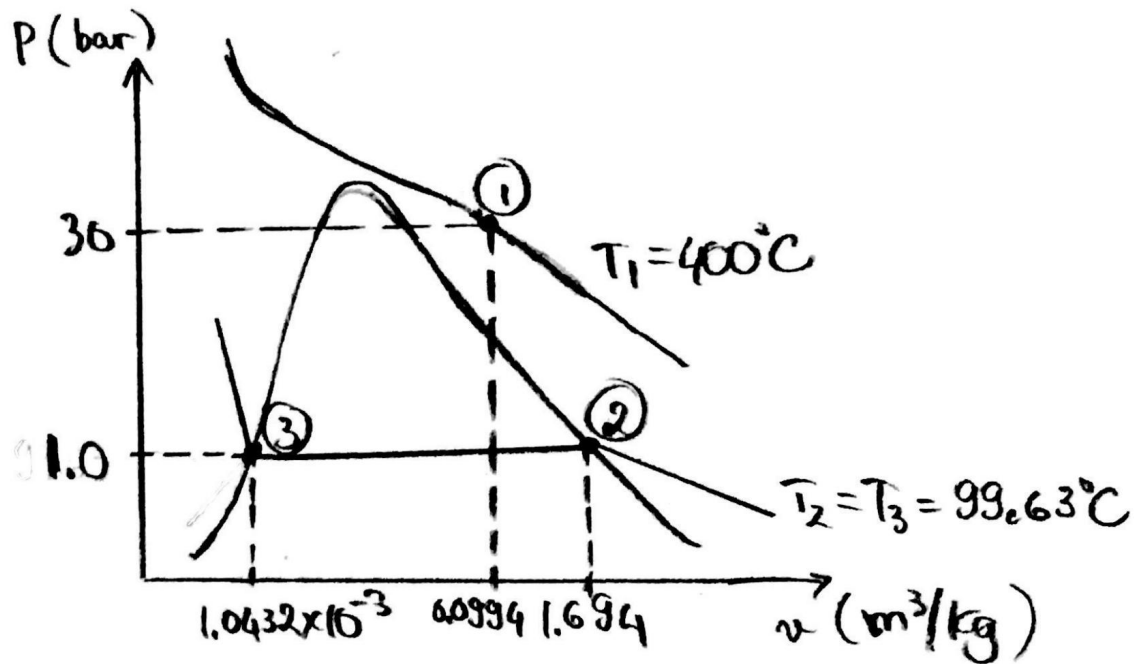
$$\dots + \left(\frac{40^2}{2} - \frac{1}{2} \left(\frac{100^2 + 90^2}{2} \right) \right) \frac{\text{m}^2}{\text{s}^2} \times \frac{1\text{ N}}{1\text{ kg m/s}^2} \times \frac{1\text{ J}}{1\text{ Nm}} \times \frac{1\text{ kJ}}{10^3\text{ J}}$$

$$= -2959.79\text{ kW} = -2.960\text{ MW}$$

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e.) $\dot{m}_i = \frac{A_i V_i}{v_i} \rightarrow A_i V_i = \dot{m}_i v_i = (2 \text{ kg/s})(0.0994 \text{ m}^3/\text{kg}) = 0.1988 \text{ m}^3/\text{s}$

f)



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